SideEffectNet – AI Powered Drug Side Effect Intelligence System

Introduction
SideEffectNet is an AI-powered healthcare intelligence system designed to analyze, retrieve, and explain drug side effects using verified medical documents. It combines NLP, Machine Learning, and Retrieval-Augmented Generation (RAG) to reduce misinformation and improve reliability.

Problem Statement
Drug side-effect information is complex, scattered, and often unreliable. Traditional chatbots may hallucinate medical information, leading to unsafe interpretations.

Proposed Solution
SideEffectNet ingests medical documents, extracts relevant side-effect information, stores it in a vector database, and generates answers grounded strictly in retrieved medical context using RAG.

Objectives
- Improve drug safety awareness
- Reduce hallucination in medical AI systems
- Demonstrate real-world AI + Healthcare application

Technologies Used
Python, NLP, Machine Learning, OCR, Vector Databases, LLMs, RAG, Streamlit/Web UI

Features
- Multi-source document ingestion
- OCR support
- AI-based side-effect analysis
- Severity categorization
- Intelligent QA chatbot

Architecture
Data Sources → Processing (OCR + NLP) → Vector Database → RAG Pipeline → Chatbot UI

Why RAG?
RAG ensures answers are grounded in verified medical documents, reducing hallucination and increasing trust in healthcare applications.

Future Scope
- Multilingual support
- Live medical API integration
- Personalized risk analysis
- Mobile application deployment

Conclusion
SideEffectNet showcases a safe, reliable, and intelligent approach to healthcare AI by combining document retrieval with controlled language generation.